

Q34 Definitions and Rules

Cal 2 Support Notes

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This reference sheet explains key terms and rules used in Question 34.

Trigonometric substitution

A method for integrals involving radicals. Common patterns:

$$\sqrt{a^2 - x^2} : x = a \sin \theta, \quad \sqrt{a^2 + x^2} : x = a \tan \theta, \quad \sqrt{x^2 - a^2} : x = a \sec \theta.$$

Example: if $\sqrt{4 - x^2}$ appears, use $a = 2$ so $x = 2 \sin \theta$.

Denominator

In a fraction $\frac{p(x)}{q(x)}$, the denominator is the bottom expression $q(x)$. In Q34, the denominator is $(4 - x^2)^{3/2}$.

Fractional exponents

$$a^{m/n} = \sqrt[n]{a^m} = \left(a^{1/n}\right)^m.$$

Example:

$$(4 - x^2)^{3/2} = \left((4 - x^2)^{1/2}\right)^3 = \left(\sqrt{4 - x^2}\right)^3.$$

Notation: $\sin^2 \theta$

$\sin^2 \theta$ means $(\sin \theta)^2$. This is standard shorthand.

Pythagorean identity

$$\sin^2 \theta + \cos^2 \theta = 1 \quad \Longleftrightarrow \quad 1 - \sin^2 \theta = \cos^2 \theta.$$

Rule: $\sqrt{ab} = \sqrt{a}\sqrt{b}$

For nonnegative a, b :

$$\sqrt{ab} = (ab)^{1/2} = a^{1/2}b^{1/2} = \sqrt{a}\sqrt{b}.$$

Inverse sine $\sin^{-1}(x)$

$\sin^{-1}(x)$ (also written $\arcsin(x)$) returns the angle whose sine is x . **Example:** if $\sin \theta = \frac{x}{2}$, then

$$\theta = \sin^{-1}\left(\frac{x}{2}\right).$$